

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0849

North Brewton 138 kV
July 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0849. The Interconnection Customer proposes a 705.7 MW natural gas combined cycle facility interconnecting at the North Brewton 138 kV substation in the Pennsylvania area. The request was received on August 16, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 388 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$232,097,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Brewton Substation-C W Burdick 138 kV	Gaylord - Consumers-N Brewton 138 kV L	218.0	246	64.3
Grahamville-C W Burdick 138 kV	N Brewton-Gaylord - Consumers Substation 138 kV	214.8	249	64.3
E Murfreesboro-Pixley 138 kV	Piney Crk Project-Caldwell 345 kV	186.3	183	5.2
Red Bluff-Gaylord - Consumers 138 kV	C W Burdick-North Brewton 138 kV	185.9	214	50.7
N Brewton-Gaylord - Consumers 138 kV	C W Burdick-N Brewton 138 kV line	179.9	223	50.7
E Murfreesboro-Little Falls 138 kV L	Caldwell-Piney Crk Project 345 kV Line	167.4	206	5.2
Gallatin Mo-Ellington 138 kV L	C W Burdick-N Brewton 138 kV L	156.2	188	49.3
Orchard Road 500/345 kV Auto	Orchard Rd-Ft Humbug 500 kV L	154.1	792	6.2
Little Falls-North Slidell 138 kV Line	Caldwell-Piney Crk Project 345 kV	152.3	230	5.2
Sierra-Miles City Substation 138 kV	Grahamville-Piney Crk Project Substation 345 kV	147.5	214	5.1
Sierra-Lava Substation 138 kV	Grahamville-Piney Crk Project 345 kV	143.6	221	5.1
Lava-Rodeo 138 kV	Piney Crk Project-Grahamville 345 kV	141.0	227	5.1
Airport-D East 1-Ellington 138 kV	C W Burdick-N Brewton 138 kV	138.9	217	49.3
Lay Dam-Rodeo 138 kV	Piney Crk Project Substation-Grahamville 345 kV	137.3	235	5.1
North Brewton-Airport-D E 1 138 kV Line	N Brewton-C W Burdick 138 kV	136.2	223	49.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Alegheny Hydro No 8 345/138kV Autotransformer	Caldwell Substation-Piney Crk Project 345 kV	126.0	389	5.8
Grahamville 345/138 kV Autotransformer	Piney Creek Project 500 kV/345 kV XFMR	123.7	353	26.6
Morganville-Ironwood Llc 345 kV	Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV Autotransformer	108.4	1,106	5.1
Collin-Potholes 138 kV	N Brewton-C W Burdick 138 kV	108.2	172	8.0
Shawville Sta 500 kV/345 kV Transformer Bank	Carbon Center Substation 500/345 kV TX	106.1	1,206	8.3
Caldwell Substation-Carbon Center 345 kV line	Blackrock-Shawville Sta 345 kV L	103.5	1,506	8.6
Sabana Grande 6501-Red Bluff 138 kV	C W Burdick-N Brewton 138 kV	103.0	211	24.3
Hilltop-Tlenenergy Martins Crk Llc Williamsport 500 kV	Hilltop 500/345kV Transformer	101.1	2,941	8.6
Hilltop-Piney Crk Project 500 kV	Caldwell-Piney Creek Project 345 kV	100.3	2,934	13.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Brewton-C W Burdick 138 kV line	Rebuild line with higher-capacity conductor	\$3,878,000	100% (\$3,878,000)
Grahamville-C W urdick 138 kV	Rebuild line with higher-capacity conductor	\$15,278,000	100% (\$15,278,000)
E Murfreesboro-Pixley 138 kV	Rebuild line with higher-capacity conductor	\$21,151,000	100% (\$21,151,000)
Red Bluff-Gaylord - Consumers 138 kV	Rebuild line with higher-capacity conductor	\$5,800,000	100% (\$5,800,000)
N Brewton-Gaylord - Consumers 138 kV	Rebuild line with higher-capacity conductor	\$8,788,000	100% (\$8,788,000)
E Murfreesboro-Little Falls 138 kV L	Rebuild line with higher-capacity conductor	\$4,099,000	100% (\$4,099,000)
Elington-Gallatin Mo 138 kV	Rebuild line with higher-capacity conductor	\$2,007,000	100% (\$2,007,000)
Orchard Rd 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$23,062,000	33% (\$7,525,000)
Ltle Falls-North Slidell 138 kV Line	Rebuild line with higher-capacity conductor	\$18,605,000	100% (\$18,605,000)
Sierra-Miles City 138 kV	Rebuild line with higher-capacity conductor	\$7,962,000	100% (\$7,962,000)
Sierra Substation-Lava 138 kV	Rebuild line with higher-capacity conductor	\$9,967,000	100% (\$9,967,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lava Substation-Rodeo 138 kV	Rebuild line with higher-capacity conductor	\$5,961,000	100% (\$5,961,000)
Airport-D E 1-Ellington 138 kV	Rebuild line with higher-capacity conductor	\$9,255,000	100% (\$9,255,000)
Lay Dam-Rodeo 138 kV	Rebuild line with higher-capacity conductor	\$4,683,000	100% (\$4,683,000)
Airport-D E 1-N Brewton 138 kV	Rebuild line with higher-capacity conductor	\$11,836,000	100% (\$11,836,000)
Allegheny Hydro No 8 Substation 345/138 kV Auto	Replace transformer with higher-rated unit	\$18,661,000	39% (\$7,342,000)
Grahamville 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$20,209,000	100% (\$20,209,000)
Morganville-Ironwood Llc 345 kV	Reconductor line	\$7,117,000	55% (\$3,898,000)
Collin-Potholes 138 kV L	Reconductor line	\$2,142,000	100% (\$2,142,000)
Shawville Sta 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$20,474,000	48% (\$9,923,000)
Caldwell Substation-Carbon Ctr 345 kV	Reconductor line	\$19,975,000	30% (\$6,009,000)
Sabana Grande 6501-Red Bluff 138 kV	Reconductor line	\$2,196,000	100% (\$2,196,000)
Hilltop-Tlenenergy Martins Crk Llc Williamsport 500 kV	Reconductor line	\$34,845,000	16% (\$5,435,000)
Hilltop-Piney Creek Project 500 kV	Reconductor line	\$203,590,000	19% (\$38,148,000)
	Total allocated to project		\$232,097,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.